

LGBM- 65 – NUTRIENTS

1. SMOTE

Average Accuracy Val K-Fold: 92.2974
Average Accuracy Train K-Fold: 99.8926
Average MCC Val K-Fold: 0.8333
Average MCC Train K-Fold: 0.9986
Average F1 Score Val K-Fold: 0.9237
Average F1 Score Train K-Fold: 0.9989
Average Precision Val K-Fold: 0.9255
Average Precision Train K-Fold: 0.9989
Average Recall Val K-Fold: 0.9230
Average Recall Train K-Fold: 0.9989
Accuracy Test: 91.41%
MCC Test: 0.8116
F1 Score Test: 0.9136
Precision Test: 0.9142
Recall Test: 0.9141

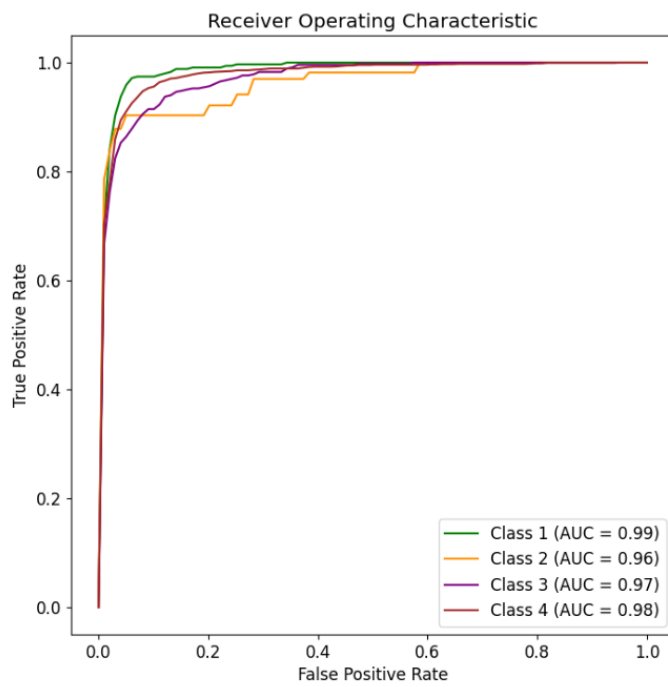
Classification Report:

	precision	recall	f1-score	support
1	0.77	0.85	0.81	68
2	0.90	0.82	0.86	11
3	0.82	0.74	0.78	93
4	0.96	0.96	0.96	422
accuracy			0.91	594
macro avg	0.86	0.84	0.85	594
weighted avg	0.91	0.91	0.91	594

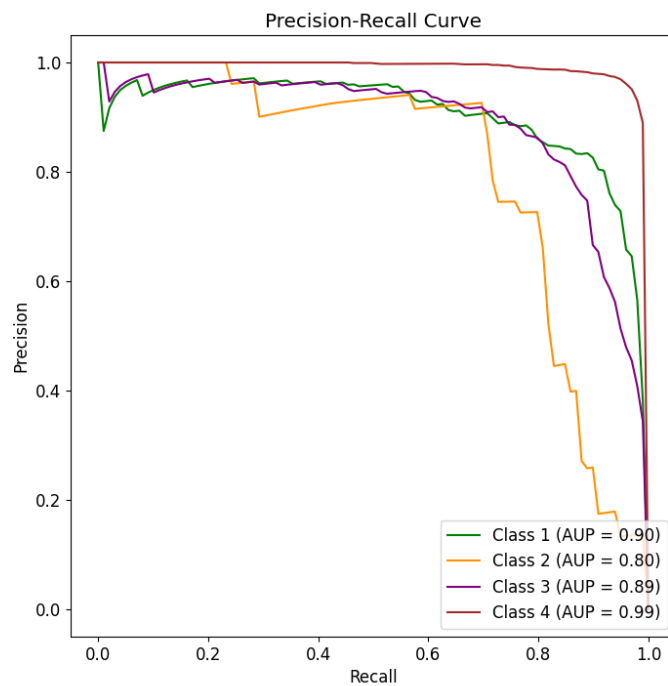
Confusion Matrix Test:

```
[[ 58  0  5  5]
 [  0  9  0  2]
 [ 13  0 69 11]
 [  4  1 10 407]]
```

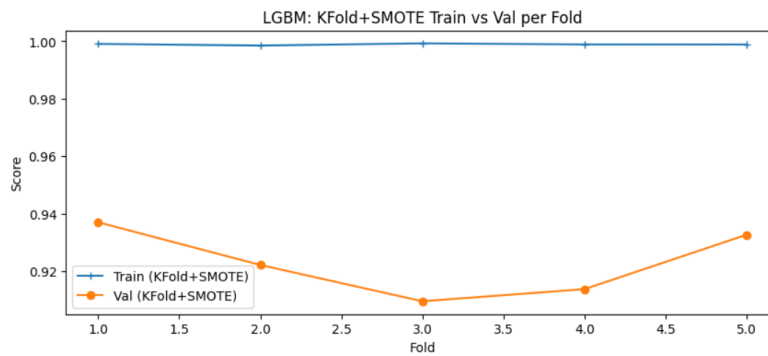
ROC curve



PR curve



Training and Validation curve



2. SMOTE + STRATIFIED

Average Accuracy Val K-Fold: 92.1286

Average Accuracy Train K-Fold: 99.8928

Average MCC Val K-Fold: 0.8309

Average MCC Train K-Fold: 0.9986

Average F1 Score Val K-Fold: 0.9212

Average F1 Score Train K-Fold: 0.9989

Average Precision Val K-Fold: 0.9246

Average Precision Train K-Fold: 0.9989

Average Recall Val K-Fold: 0.9213

Average Recall Train K-Fold: 0.9989

Accuracy Test: 90.24%

MCC Test: 0.7877

F1 Score Test: 0.9026

Precision Test: 0.9036

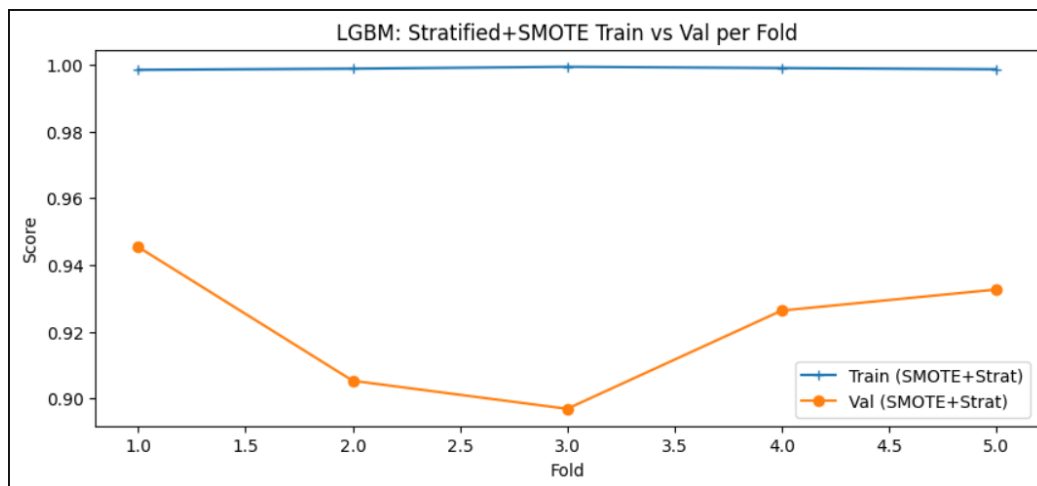
Recall Test: 0.9024

Classification Report:

	precision	recall	f1-score	support
1	0.75	0.84	0.79	68
2	0.90	0.82	0.86	11
3	0.78	0.74	0.76	93
4	0.95	0.95	0.95	422
accuracy			0.90	594
macro avg	0.85	0.84	0.84	594
weighted avg	0.90	0.90	0.90	594

Confusion Matrix Test:

```
[[ 57  0  6  5]
 [  0  9  0  2]
 [ 12  0 69 12]
 [  7  1 13 401]]
```



3.STRATIFIED

Average Accuracy Val K-Fold: 91.2455
 Average Accuracy Train K-Fold: 99.6949
 Average MCC Val K-Fold: 0.8047
 Average MCC Train K-Fold: 0.9933
 Average F1 Score Val K-Fold: 0.9102
 Average F1 Score Train K-Fold: 0.9969
 Average Precision Val K-Fold: 0.9115
 Average Precision Train K-Fold: 0.9970
 Average Recall Val K-Fold: 0.9125
 Average Recall Train K-Fold: 0.9969

Accuracy Test: 91.25%

MCC Test: 0.8042

F1 Score Test: 0.9100

Precision Test: 0.9094

Recall Test: 0.9125

Classification Report:

	precision	recall	f1-score	support
1	0.79	0.79	0.79	68
2	0.90	0.82	0.86	11
3	0.81	0.70	0.75	93
4	0.95	0.98	0.97	422
accuracy			0.91	594
macro avg	0.86	0.82	0.84	594
weighted avg	0.91	0.91	0.91	594

Confusion Matrix Test:

```

[[ 54  0 11  3]
 [  0  9  0  2]
 [ 11  0 65 17]
 [  3  1  4 14]]

```

